Just so, if we are confident that there are a huge number of different universes, we should not take the fact that we are in a life-permitting one to be evidence for much of anything.

So the key question is: do we have good reason to think that the multiverse hypothesis is true?

A first point to note: it would be very surprising if this hypothesis were true. For, if it is, there are very many — perhaps infinitely many — other universes, each as real as ours, in which some near-duplicate of you exists. There is, for example, very likely one in which there exists some being with a qualitatively identical history to you who differs from you only in that she or he scratched his nose one second ago.

This does not show that the multiverse hypothesis is false; the universe might be strange, and science has often shown us that it is. But it does suggest that the multiverse hypothesis is not one that we should believe without argument.

But one might think that the very facts used in the fine-tuning argument can be used to support the multiverse hypothesis. For consider the following argument:

But, while this reasoning sounds plausible, consideration of parallel cases shows that something has gone wrong.

A Bayesian argument for the multiverse

It is very, very improbable that our universe is the only one and, just by chance, the constants came to be set in such a way as to make life possible. But if there were many many universes, it would not be very improbable that one would be life supporting. So, the fact that our universe is life-supporting is strong evidence in favor of the multiverse hypothesis.

fine-
tuning of the
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Bayes' theorem

the argument

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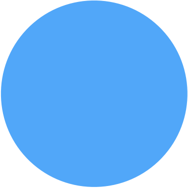

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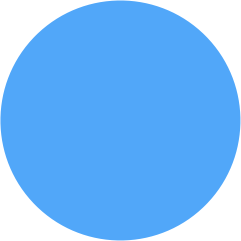

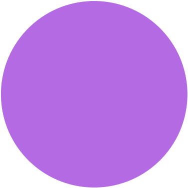

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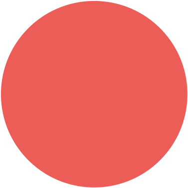

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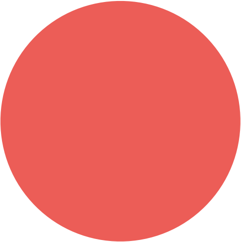

Enough about probability (for now). Let's turn now to some results from follows.

contemporary physics which will be important to the argument which

A first question: what does the theory provided by physics include?

‘‘The standard model of physics presents a theory of the electromagnetic, weak, and strong forces, and a classification of all known elementary particles. The standard model specifies numerous physical laws, but that's not all it does. According to the standard model there are roughly two dozen dimensionless constants that characterize fundamental physical quantities.’’ (Hawthorne & Isaacs, ‘‘Fine-tuning fine-tuning’’)

These “dimensionless constants” will be our focus.

One such constant is the cosmological constant, which measures the energy density of empty space.

could have different values consistent with the laws of physics.

The cosmological constant is just a number which (as far as we know) likely it is, given the laws of the nature, that the fundamental constants (like the cosmological constant) would fall in a certain range.

One thing that the standard model of physics gives us is a measure of how hard to see how life could have evolved. If there were no planets, it is hard life to have evolved. For example, if there were nothing but hydrogen, it is to see how life could have evolved.

We can make certain plausible assumptions about what it would take for constant has a value which would permit the evolution of life.

Given assumptions such as these, we can look at what the standard model of physics tells us about how likely it is that, for example, the cosmological

“Physicists have determined the (approximate) values of the fundamental constants by measurement. (There's no way to derive the values of the fundamental constants from other aspects of the standard model. Any quantities that could be so derived wouldn't be fundamental.) Still, the underlying theory favored some sorts of parameter-values over others. ... Physicists made the startling discovery that -- given antecedently plausibly assumptions about the nature of the physical world -- the probability that a universe with general laws like ours would be habitable was staggeringly low.”

supporting value, given the laws of nature, is very low.

This is the claim that that the cosmological constant having a life-

We should emphasize just how small we take the life-permitting parameter values to be according to the physically-respectable measures. “Small” here doesn’t mean “1 in 10,000” or “1 in 1,000,000”. It means the kind of fraction that one would resort to exponents to describe, as in “1 in 10 to the 120”. The kind of package that we have in mind tells us that only a fantastically small range is life permitting.

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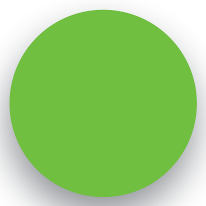

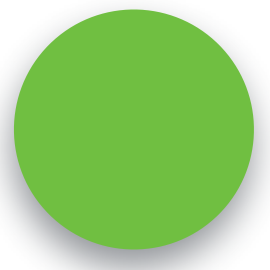

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probability theory into an argument for the existence of God.

Let's now ask how to turn these facts about current physics and of the universe.

in the urn, so we can consider two different hypotheses about the origins

Much as we considered two different hypotheses about the number of balls

Let the **design hypothesis** be the hypothesis that the universe was hypothesis that it was not.

created by an intelligent designer. Let the **non-design** hypotheses be the

Let our **evidence** (e) be the fact that the cosmological constant has a value in the life-permitting range.

Then what do we know about the relevant probabilities?

$$Pr(e \mid \text{non-design}) = \frac{1}{10^{120}}$$

$$Pr(e \,|\, \mathsf{design}) = 0.5$$

Bayes' theorem

$$P(h|e) = \frac{P(h) * P(e|h)}{P(e)}$$

design hypothesis given our evidence is a matter of just plugging in the

With this information in hand, figuring out the probability of the nonnumbers.

$$Pr(\text{non-design}) = 0.5$$

$$Pr(\text{design}) = 0.5$$

$$Pr(e) = 0.25$$

$$Pr(\text{non-design} \mid e) = \frac{0.5 * \frac{1}{10^{120}}}{0.25} = \frac{2}{10^{120}}$$

consider the odds of winning Powerball one trillion times in a row. Call that a “super

Powerball.”

to give you some idea: the odds of winning Powerball are about 1 in 300 million. Now

It is difficult to think about numbers as large as the denominator of this fraction. But

Now consider the odds of winning a super Powerball one trillion times in a row. Call that a "super duper Powerball."

Now consider the odds of winning a super duper Powerball one trillion times in a row.

odds of the universe being life-permitting by chance.

The odds of this happening are about $1 / 10^{44}$ — so much, much higher than the probability of 0.5 to the non-design hypothesis, you should think that the chances of

This means that if you simply take the physics at face value, and begin by assigning a the non-design hypothesis being true are vastly lower than the chances of winning a super duper powerball a trillion times in a row.

By contrast, the probability of the design hypothesis is very close to 1. It is, in fact, 1 minus the very small number we were just discussing:

$$Pr(\text{design} \mid e) = 1 - \frac{2}{10^{120}}$$

This is about as close to certainty as it is possible to get.

objections

1. P

2. P

3. P(design)=P(non-design)=0.5.

5. P

4. Bayes' theorem.

. (1,2,3,4)

8. If the universe was created by an

7. P(design) ≈ 1. (5,6)

6. Life exists.

intelligent designer, then God exists.

C. $P(\text{God exists}) \approx 1$. (7,8)

$$(\texttt{life}|\texttt{design}) = 0.5$$

$$(\texttt{life}|\texttt{non-design}) = \frac{1}{10^{120}}$$

$$(\texttt{design}|\texttt{life}) \approx 1$$

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